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Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Education & Outreach Lead		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
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No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is 11/30/2006, per info from PI office (2/2009)		
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COI Name (Institution):	Hains, Julia (Purdue University) Banks, Kathy (Purdue University)		
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	The NSCORT Education and Outreach provides an avenue to engage and educate higher education students, K-12 educators/students and the general public in the ALS/NSCORT center's investigations of the synergistic concepts and principles required for regenerative life-support in extended-duration space exploration. The Education and Outreach program focuses on two thrust areas: Technical/Higher Education which includes the Summer Fellowship Program and ALS/NSCORT Distance Learning Course and 2) K-12 Education which includes Key Learning Collaborative Project, Mission To Mars and Equivalent Systems Mass Analysis of Plant Growth. Technical/Higher Education has grown significantly. The Summer Fellowship Program trained 7 undergraduate fellows in the Summer 2004. Included in the fellowship were two elementary education students from Alabama A& M who		

Task Description:	received intensive training on the K-12 programs of Education and Outreach. Funding was allocated to employ these students during the 2004-2005 school year in Alabama, extending the reach of NSCORT into the Alabama schools. ALS/NSCORT Distance Learning Course was offered for the first time in the Spring 2004. Thirty students were enrolled in the course. K-12 Education grew exponentially. Key Learning Collaborative Project developed significant research programs on-going in the school laboratory. The project was expanded to include an "Explore Mars" summer camp which provided 150 Key Learning students the opportunity to spend 3 days/nights on the Purdue University campus, interacting with NSCORT researchers and graduate students. The camp was financed with funds of over $20,000.00 which were obtained through grants written by NSCORT Outreach manager. Lafayette School Corporation funded ($4,000) and piloted the Mission To Mars program in four elementary schools in 2004. Numerous professional development programs were offered throughout Indiana as well. Two significant proposals have been submitted to date for the funding of professional development workshops and nationwide dissemination of the Mission To Mars program. Equivalent Systems Mass (ESM) Analysis of Plant Growth project was funded by a $25,000 SBIR grant to develop an 18 week curriculum module for high schools involved with Project Lead the Way. This module incorporates BPES growing systems and students' analysis of calculated ESM values. A pilot program is on-going to date with an expanded pilot program planned for January 2005. Project Lead The Way will incorporate this module into their national network in June 2005. This year, ALS/NSCORT Education and Outreach K-12 programs alone have reached 516 teachers and 16,177 students. Julia Hains-Allen Education and Outreach • Increase the number of students/educators who are involved in NASA related education opportunities. • Increase the scientific literacy of K-12 educators, students and general public, fostering public awareness and the earth benefits of NASA research. • Increase the number of students directly engaged in NASA research. • Increase the number and diversity of students who pursue graduate degrees and research careers in NASA related science, engineering and technology fields. • Increase higher education research capability and opportunities that attract and prepare students for NASA related careers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	• Increase the number of students/educators who are involved in NASA related education opportunities. • Increase the scientific literacy of K-12 educators, students and general public, fostering public awareness and the earth benefits of NASA research. • Increase the number of students directly engaged in NASA research. • Increase the number and diversity of students who pursue graduate degrees and research careers in NASA related science, engineering and technology fields. • Increase higher education research capability and opportunities that attract and prepare students for NASA related careers.
	Cumulative Research Progress to Date Education and Outreach program focuses on two major areas: 1) Technical/Higher education, and 2) K-12 education. Technical/Higher Education Summer Fellowship Research Program: This program offers undergraduate students from Alabama A&M, Howard and Purdue universities eight weeks of research experience under the direction of an ALS/NSCORT researcher. Summer Fellowship applicants indicate a preference for a particular faculty member and NSCORT site. Fellowships are awarded according to faculty choice and project compatibility. ALS/NSCORT provides round trip transportation, room at the research institution plus a $3600 stipend for the eight-week research opportunity. A two-day scientific symposium culminating the summer research experience is held at one of the research institutions in August, allowing each summer fellow to present their work to their peers in a scientific presentation. In addition to the data gathered from the 2003 program, the ALS/NSCORT Education and Outreach Group consulted numerous sources on "best practices" for summer fellowship programs to gather information on redesigning portions of the program. The following changes were made to the 2004 Summer Fellowship Program: 1. Funding included room and transportation to/from the research institution. 2. Fellowships were offered in Education/Outreach increasing the geographical contact area for outreach. Only Alabama A&M and Howard students were eligible. Summer Fellowship Undergraduate Students SURF Students In 2004, seven fellowships were offered. Four students from Howard University and three students from Alabama A&M participated in the program. Two of the students from Alabama A&M received fellowships in Education/Outreach, creating a new program within the fellowship program. In addition to the 8 week summer research, this new addition allows the Alabama students to continue the fellowship and present Education/Outreach programs in Alabama during the 2004-2005 school year. The program will significantly increase the reach of ALS/NSCORT Outreach into the Alabama educational community. Space Advanced Life Support Distance Course (see James Alleman's section for report below) K-12 Education Key Learning Community Project In the fall of 2003, the ALS/NSCORT Center and Key Learning Community in Indianapolis formed a coalition to enhance the learning of students in Indianapolis. The coalition has designed a rigorous course of action to address the problem that many underrepresented students in Indianapolis are not pursuing careers in science, technology and engineering. Key Learning Community Collaborative Project provides exposure, mentoring and research opportunities for 9-12th grade students at Key Learning Community. The collaboration will focus on developing successful 9th-12th grade outreach strategies for ALS/NSCORT. In a science laboratory at Key Learning is an actual prototype bioreactor designed by Purdue Engineering researchers to remove surfactants from gray water in the Mars habitat. Key students are performing research on this bioreactor, including reactor performance assessment with liquid influent. In addition, Key students are raising Tilapia, the fish that will serve as a bioreactor of nutrient species thus reducing Equivalent Systems Mass (ESM) in the ALS/NSCORT space habitat design. Hydroponics plant growth research was added in July 2004. Purdue University researchers and graduate students provide mentoring for the students during the research period via email communication, classroom visits, university trips and a summer camp experience. This collaborative project provides Key Learning Community teachers and students with an arsenal of experiences that make learning science, technology, and engineering exciting without sacrificing rigor.

Task Progress:

Explore Mars Camp Expanding the collaborative relationship, a summer camp experience was added in August 2004. One hundred and fifty Key Learning high school students stayed on Purdue's campus from August 4th to August 6th. The camp experience provided some students with an expanded opportunity to conduct research in other areas of ALS/NSCORT while other students will increase the depth of the present research on-going in the school. Further ensuring the success of the camp program, students self-selected the research area in which they wished to work during the camp experience. Key students were also exposed to other areas of engineering provided by a new partner, Department of Engineering Education, on Purdue's campus. This cutting-edge department ran short courses for the Key students during camp, focusing on topics such as Nosecone Simulation, Radio-Controlled Engines, Space Materials and Engineering a Sneaker. An important component of the camp experience was the opportunity for some students to create and produce a documentary on Explore Mars and the on-going research. A select group Key Learning communication students worked on this product in lieu of the laboratory research. The result of this will be a 10 minute video that can be used by ALS/NSCORT outreach in the future, showcasing the collaborative program. Over $20,000 in funds from partners were leveraged for this camp experience. In addition, the Explore Mars Camp integrated the Summer Fellowship program into the camp experience. Undergraduate students participating in the Summer Fellowship Program served as mentors and counselors in the dorms while graduate students and researchers advised students in the laboratory research.

Mission to Mars Education and Outreach has developed and pilot tested a standards/inquiry-based module, introducing 5th-8th grade students to the complex issues involved with living in a habitat on Mars. The module includes laboratory exercises revolving around the study of plant growth, ecosystems, water and waste treatment, recycling and food production. The "Mission to Mars" module provides teachers with a collection of standards-based, multi-disciplinary hands-on activities that are easily used in classrooms and after school programs. In keeping with the National Science Education Standards (NRC, 1996) and Inquiry in the National Science Education Standards (NRC, 2000), the "Mission to Mars" module facilitates science instruction that actively involves students in meaningful experiences that provide opportunities to develop inquiry abilities and understanding to promote scientific literacy. Module activities are based on cutting-edge research that is ongoing in the ALS/NSCORT. These direct links with research and current events serve to build teacher expertise in cutting edge science, while stimulating student interest by connecting learning to real-life challenges. The "Mission to Mars" module includes the background information teachers need to understand the science content, eleven comprehensive 5-8th grade standards-based activities, teacher guides and material lists along with implementation instructions. Macon Fish, ALS/NSCORT graduate student, has developed a content and attitudinal assessment tool which is being piloted in 15 classrooms during October-December 2004. Formal (1-2 day) professional development programs in FY04 trained one hundred and twenty three (123) 5th-8th grade teachers and Extension educators. Those formal professional development programs are indicated by an asterisk in the Projects/ Presentation section of this report. Evaluation materials have been given to all educators participating in professional development programs to date. Data obtained from the evaluation clearly demonstrates that the "Mission To Mars" program is: a) highly applicable to their classroom and b) contains a large amount of standards/based science content they can easily integrate into their curriculum.

Response to this program has been overwhelming. The pilot program in FY04 resulted in the participation of 100 Lafayette School Corporation students. In FY05, Lafayette School Corporation has requested the program be offered to 250 students, expanding the program from after-school to intersession classrooms. Funds of $8,000 from LSC will be used for this request. A professional development program offered in the fall 2004 was filled the first week following an email solicitation. 15 teachers are on the waiting list for the next professional development program.

Project Lead The Way NASA Advanced Life Support has joined with Orbital Technologies to design Biomass Production Educational Systems (BPES), a research grade plant growth chamber that can be used in high school classrooms across the country. ALS/NSCORT Education and Outreach, lead by Julia Hains-Allen, has designed learning modules integrating the BPES and ESM analysis. Students will use the research grade plant growth chambers and design experiments that investigate: • Variables needed for plant health, determining best conditions for optimum growth • Species and varieties best suited for space travel • Design improvements to plant growth chambers Using experimental results from the BPES and statistical analysis, students calculate ESM values on all variables, species and design improvements. Data provided by NASA allows students to compare their experimental results (ESM values) and determine the feasibility of their experimental designs. A website will be developed, serving as a student depository of data, student resource, and student collaboration center. This is a collaborative project between the ALS/NSCORT team and Project Lead The Way. The ALS/NSCORT team will be responsible for the development/revisions of the learning module and Project Lead The Way will be responsible for the implementation and dissemination of the module within their national network. Background In February 2004, Project Lead The Way (PLTW) contacted Julia Hains-Allen for assistance in the development of a new sequence of learning modules for Biotechnical Engineering. In response to the request, a team consisting of Julia Hains-Allen, Gus Koerner, Ross Remiker, Gioia Massa, George Applequist and Macon Fish designed a skeleton framework for a module Equivalent Systems Mass Analysis of Plant Growth. This module was presented to Indiana Department of Education Project Lead The Way team in March and was accepted into the Project Lead The Way National Biotechnical Engineering sequence. Project Lead The Way Inc. (PLTW) is a national program forming partnerships among Public Schools, Higher Education Institutions and the Private Sector to increase the quantity and quality of engineers and engineering technologists graduating from our educational system. Early on, PLTW became partners with the High Schools That Work initiative of the Southern Regional Educational Board (SREB) with schools in 27 states. Currently, PLTW offers its program in 31 states from New York to California. Module Development • June 1, 2004 Orbital Technologies delivered three BPES systems to ALS/NSCORT for the initial development of the learning module Equivalent Systems Mass Analysis of Plant Growth. Macon Fish, Agriculture Education graduate student, lead a team in the development of the learning module. The team consisted of Macon Fish, Julia Hains-Allen, Jeff Martin, Project Lead The Way teacher in Indianapolis and Carol Martin, Math teacher in Lakeland, FL. Funding for the module development was provided by a $25,000 Orbitec sub-contract via a NASA SBIR grant. Key concepts addressed in the module are: • Factors that effect plant health and optimum growth • Analysis factors required for ESM value calculations and systems applications • Plant varieties and species to be used in space travel • Statistical analysis of data • Investigating unprecedented problems Benefits of space research to Earth agriculture systemsAugust 1, 2004 module development was completed and the initial pilot testing began at McKenzie Learning Center, MSD Washington Township School Corporation, Indianapolis. This pilot is lead by Jeff Martin, Project Lead The Way teacher at McKenzie. A second pilot testing period is planned in January 2005. This testing period will include 10 teachers throughout Indiana and Florida. At the completion of this pilot period, the learning module will be integrated into the Project Lead The Way curriculum. Indiana Department of Education Project Lead The Way will be financially responsible for the professional development and initial implementation into the Project Lead The Way national network. National implementation will be the financial responsibility of Project Lead the Way national

	organization. ALS/NSCORT team will serve as consultants and resources during the first and second pilot period. Julia Hains-Allen will serve on the Project Lead the Way team as the lead consultant and liaison between the ALS/NSCORT development team and Project Lead the Way National throughout the entire module testing, implementation, professional development, and subsequent national dissemination to ensure NASA missions and goals are met. 'Space Advanced Life Support' Distance Course : Jim Alleman Our Spring 2004 ALS/NSCORT Distance Learning 'Space Advanced Life Support' course was developed and presented on an experimental basis to provide a multi-state, multi-disciplinary, and multi-speaker opportunity for ALS/NSCORT faculty members to mutually build upon and reinforce their expertise in the area of advanced life support, while transferring this knowledge on a credit (3-hr) earning basis to graduate and undergraduate students either directly working with, or academically curious about, our ALS research mission. The course was managed by Jim Alleman on a Tues-Thurs 75-min schedule basis, with 15 total guest speakers covering various classes during the entire semester's 30-class schedule. Student enrollment included 26 local students and 5 remote students, and another 4-5 NSCORT staff typically listened in via the Web during most classes. The volunteer speaker roster was largely composed of NSCORT faculty, as well as doctoral-level graduate students and NSCORT center personnel and post-doctoral research assistants. In addition, the class greatly enjoyed lectures by former and current NASA personnel, including astronaut Guy Gardner (STS-27 and STS-35) and Karen Ross (the Manager for Space Food Processing with United Space Alliance), and a number of field tours to ongoing NSCORT project sites. Both real-time local and remote, plus post-presentation video, access to the course lectures was provided using state-of-the-art Web-based systems. In general, the content of the course offered a state-of-the-art introduction to the complex challenge of life support for astronauts working in space (i.e., during lunar, Mars, etc. missions) covering each of the major interdisciplinary aspects. As can be seen in reviewing the syllabus given on the following page, the following interdisciplinary aspects were addressed: o water supply and closed-loop recycle o waste processing and high-level water/nutrient recovery o air rejuvenation and bioregenerative oxygen resupply o water, waste, and air disinfection o food packaging and preservation o bioassay assessment of critical life support via molecular biology tools o energy-efficient crop production In retrospect, this course provided yet another creative opportunity for ALS/NSCORT researchers to participate in, and contribute to, a team-building process that synergistically contributes to our center's success as a whole. Yet another benefit to the course.
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Presentation	Hains-Allen, Julia "Explore Mars Camp"" Key Learning Community Aug-2004
Presentation	Hains-Allen, Julia, Fish, Macon "Mission To Mars" Challenger Learning Center Teachers Aug-2004
Presentation	Hains-Allen, Julia, Fish, Macon, Martin, Jeff, "Project Lead The Way Pilot" August-December 2004 Aug-2004
Presentation	Hains-Allen, Julia, Quimpo, Robert, "Mission To Mars" After-school/intersession program September 2004-April 2005 Sep-2004
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